

SDG 2: Zero Hunger Food Inventory and Distribution Management System

Submitted by:
Ariong, Ali Jr.
Bigcas, Paul Sander
Dela Cruz, Kurt Ahron
Virayo, Charo

Submitted to:
Ms. Justin Louise R. Neypes

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PROJECT OVERVIEW

The SDG 2: Zero Hunger Food Inventory and Distribution Management System is a desktop-based application developed using [VB.NET](#) and SQL Server. The main objective of the system is to help organizations properly manage donated food items, monitor inventory stocks, distribute food supplies, and generate reports efficiently.

Many organizations still use manual methods such as paper records and handwritten logs when managing food donations and distributions. Because of this, records become inaccurate, reports may be delayed, and food items may expire without being noticed. This can lead to food wastage and poor distribution management.

To solve these problems, the researchers developed a computerized system that automates inventory monitoring, donating recording, and food distribution. The system also helps monitor expiration dates to reduce food waste and improve the organization of food resources.

The system includes the following major functions:

- User Login and Role Management
- Inventory Management
- Food Distribution
- Donation Logging
- Report Generation
- Expiration Monitoring

Through this project, organizations can manage food supplies more efficiently and provide better service to beneficiaries and communities.

Chosen UN Sustainable Development Goal (SDG)

SDG 2: Zero Hunger

This project supports the United Nations Sustainable Development Goal 2, also known as “Zero Hunger.” The goal focuses on ending hunger, improving food security, and promoting proper nutrition for all people.

The system contributes to this goal by helping organizations properly track and distribute donated food supplies. Since the system monitors inventory and expiration dates, it helps reduce food waste while ensuring that food reaches beneficiaries on time.

The project specifically supports the following SDG 2 target:

“End hunger, achieve food security, improve nutrition, and promote sustainable agriculture.”

By improving food inventory and distribution processes, the system promotes better food resource management within the community.

Requirements Analysis

Functional Requirements (What the system does)

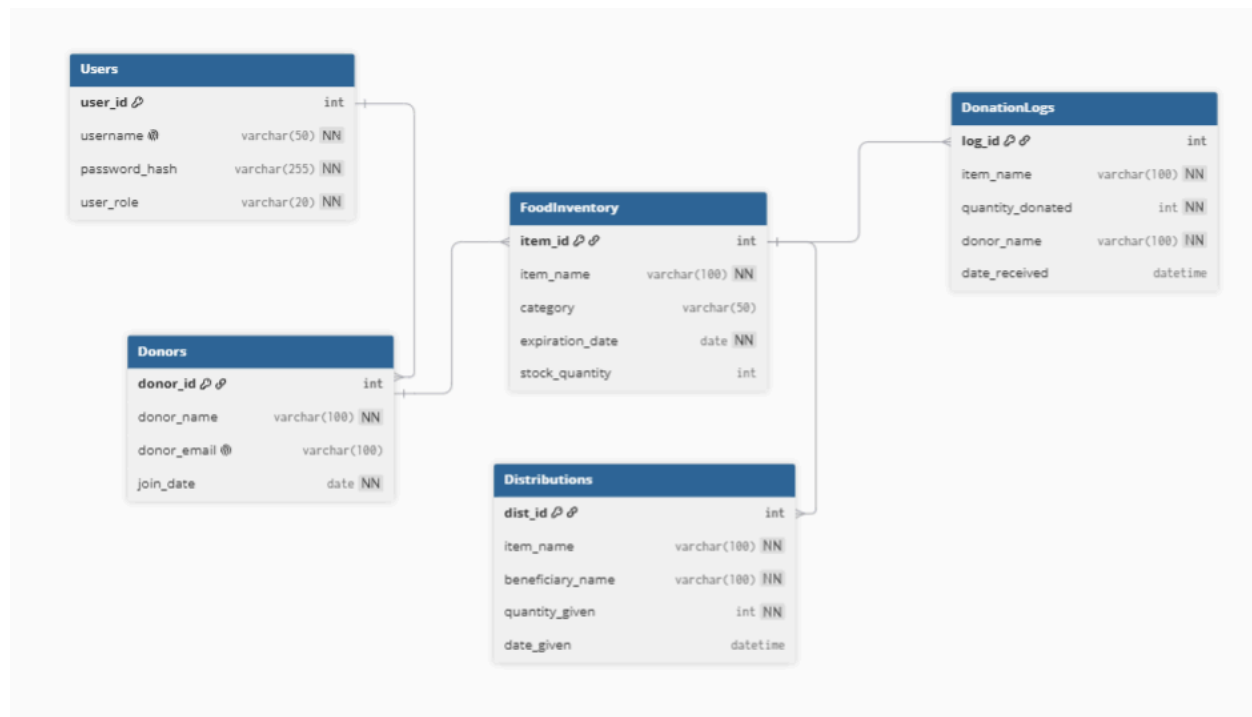
- FR-1: User Login & Roles: Users (Donors, Inventory Managers, and NGO Coordinators) must be able to log in securely and only see the tools and data assigned to their specific role
- FR-2: Inventory Management: Staff must be able to easily add, view, update, and remove food items in the stock database (Full CRUD functionality).
- FR-3: Automatic Expiration Alerts: The system needs to constantly scan the database and automatically highlight any food items whose expiration_date falls within a critical window (e.g., expiring within 7 days).
- FR-4: Donation Dispatch & Matching: The system must help coordinators match available, expiring food batches with active food requests submitted by nearby shelters.
- FR-5: Reporting Dashboard: The system must generate clean, everyday reports tracking stock levels, waste prevented, and total meals successfully distributed.

Non-Functional Requirements (How the system performs)

- NFR-1: Data Integrity: The database must enforce strict rules (such as blocking empty/null entries for names and dates) to prevent messy, broken, or orphaned records.

- NFR-2: System Stability: The program must handle data anomalies gracefully. If it runs into a data type mismatch, it should throw a predictable error instead of crashing the entire application.
- NFR-3: Speed & Performance: Queries looking up upcoming expirations must load quickly taking less than 500 milliseconds so users aren't left waiting on slow screens.
- NFR-4: Clean Code & Maintainability: The backend code should be split logically into partial classes. This ensures that if the database schema is updated or regenerated, any custom code written by our developers won't get overwritten.

Entity Relationship Diagram (ERD)



Data Dictionary

Table 1: Users

Stores account credentials and access permissions for anyone using the software.

Field Name	Data Type	Constraints/ Properties	Description
user_id	INT	IDENTITY(1,1), Primary Key	Auto-incremented system identification code.
username	VARCHAR(50)	Unique, NOT NULL	Account access identification name (Login).
password_hash	VARCHAR(255)	NOT NULL	Secured string containing account password.
user_role	VARCHAR(20)	NOT NULL	Assigned role profile level (Standard User / Admin).

- **Initial Seed Data:** Includes ('staff', 'staff123', 'Standard User') and ('admin', 'admin123', 'Admin').

Table 2: Donors

Maintains profile indexes for corporate and private external food suppliers.

Field Name	Data Type	Constraints/ Properties	Description
donor_id	INT	IDENTITY(1,1), Primary Key	Unique identification number for the food supplier.
donor_name	VARCHAR(100)	NOT NULL	Registered name of individual or business donor.
donor_email	VARCHAR(100)	Unique	Monitored digital

			mailing address profile line.
join_date	DATE	NOT NULL	Calendar date the provider entered the system.

Table 3: FoodInventory

racks individual operational product quantities and safety dates inside active storage.

Field Name	Data Type	Constraints/ Properties	Description
item_id	INT	IDENTITY(1,1), Primary Key	Unique item track value generated automatically.
item_name	VARCHAR(100)	NOT NULL	Descriptive title of the food item (e.g., "Canned Beans").
category	VARCHAR(50)	Allow NULL	Food department sorting group identifier.
expiration_date	DATE	NOT NULL	Critical product safety/spoiled baseline date.
stock_quantity	INT	Allow NULL, Check: >= 0	Current on-hand quantity balance units. Linked to constraint chk_positive_stock.

Table 4: DonationsLogs

An immutable tracking history recording every inbound donation transaction.

Field Name	Data Type	Constraints/ Properties	Description
log_id	INT	IDENTITY(1,1), Primary Key	Auto-incrementing logging line unique index key.
item_name	VARCHAR(100)	NOT NULL	Title of product handled during intake process.
quantity_donated	INT	NOT NULL	Gross inventory volume added to warehouse stock.
donor_name	VARCHAR(100)	NOT NULL	Identity name tracking back to donor source.
date_received	DATETIME	Default: GETDATE()	Chronological transaction server input timestamp.

Table 5: Distributions

Tracks outgoing delivery tracking data out to partner community distribution destinations.

Field Name	Data Type	Constraints/ Properties	Description
dist_id	INT	IDENTITY(1,1), Primary Key	System shipment record index number.
item_name	VARCHAR(100)	NOT NULL	Product nomenclature reference description.
beneficiary_name	VARCHAR(100)	NOT NULL	Title or group reference of receiving organization.

quantity_given	INT	NOT NULL	Item count value deducted from storage reserves.
date_given	DATETIME	Default: GETDATE ()	Complete calendar tracking stamp for the shipment.

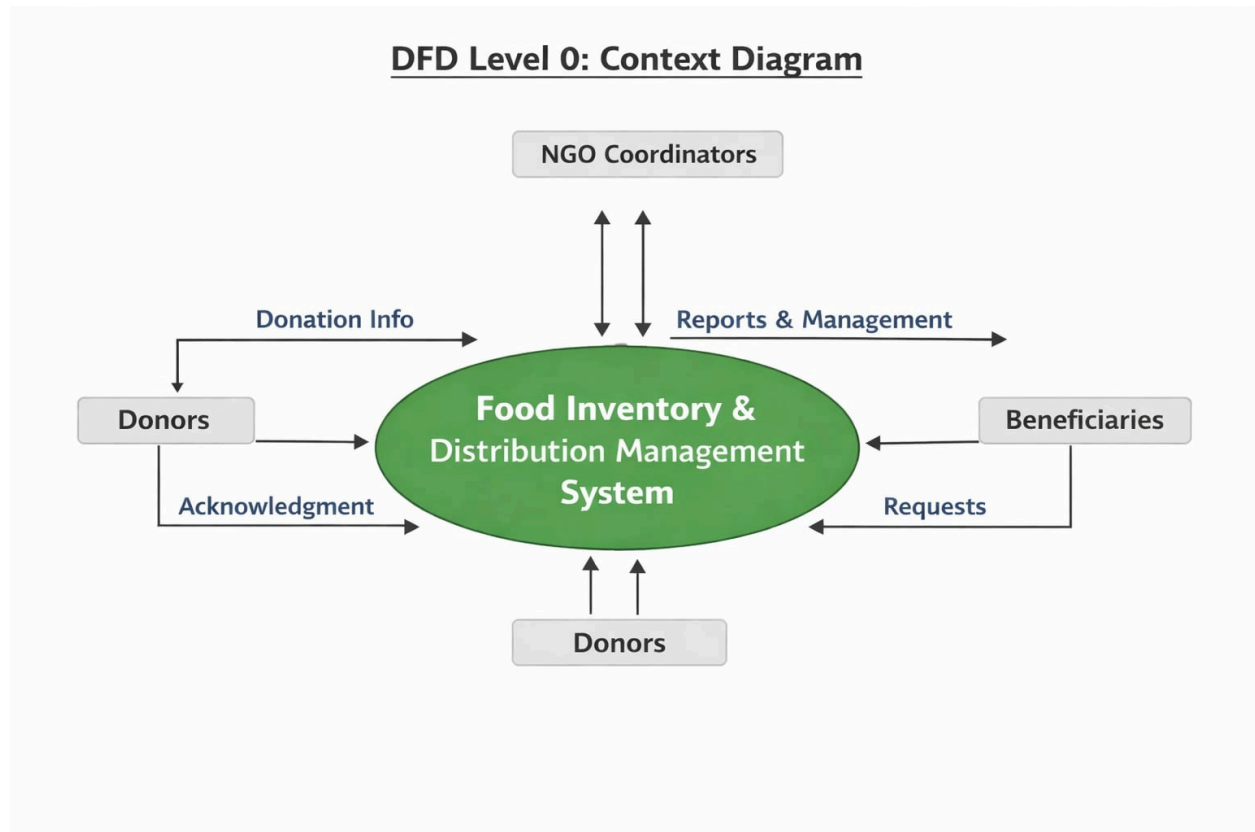
Virtual Structure: View_ExpiringStock

Virtual framework monitoring impending distribution inventory windows.

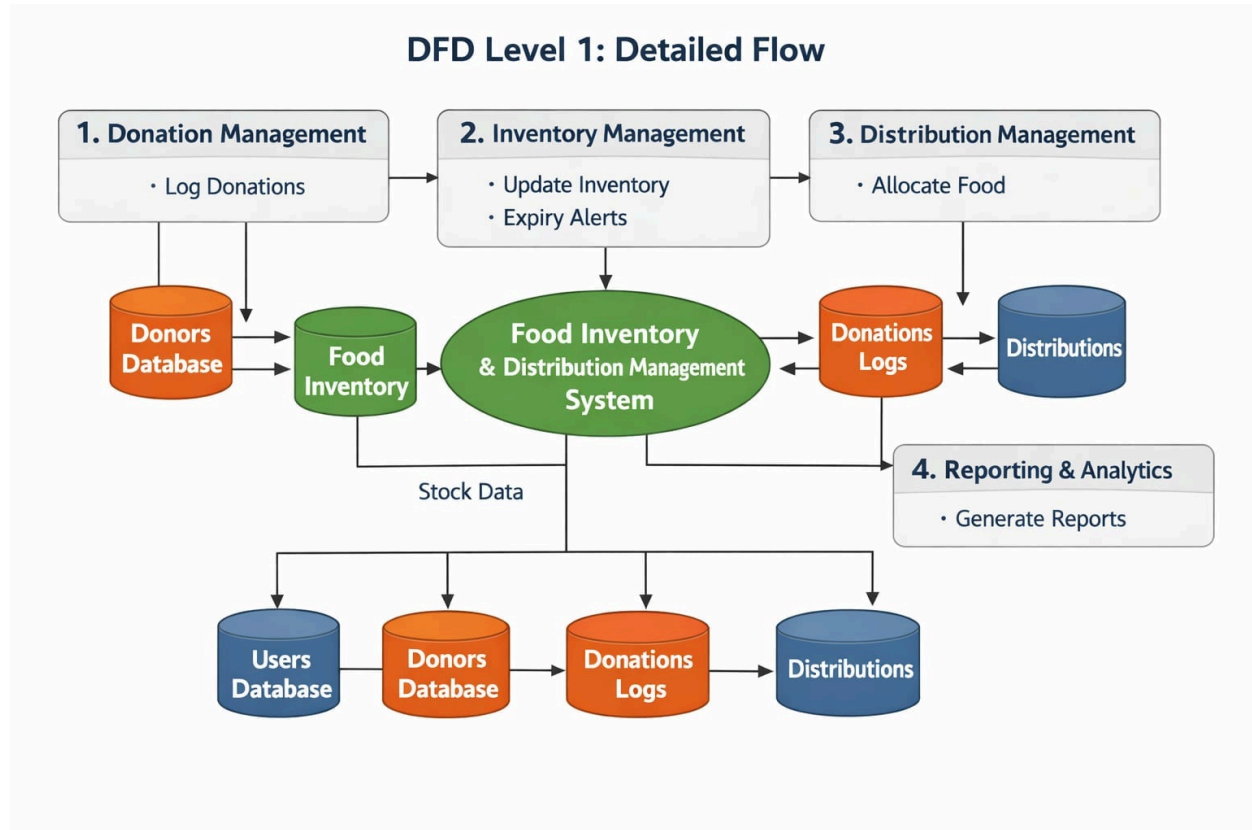
```
CREATE VIEW View_ExpiringStock AS
SELECT
    item_name,
    expiration_date,
    stock_quantity
FROM
    Inventory
WHERE
    DATEDIFF(day, GETDATE(), expiration_date) < 30;
GO
```


System Architecture (Data Flow Diagrams)

DFD LEVEL 0



DFD LEVEL 1



Individual Contributions Matrix

Group Members	Contribution
Ali Jr. Ariong	User Login & Role Management
Charo Virayo	Donation & Distribution Module
Paul Sander Bigcas	Reports, Documentation and System Architecture
Kurt Ahron Dela Cruz	Inventory Management Module

1. **Introduction:** Project Overview and chosen UN SDG Target.
2. **Requirements Analysis:** Full list of Functional and Non-Functional requirements.
3. **Database Design:**
 - Entity Relationship Diagram (ERD).
 - Data Dictionary (Table names, fields, types, and constraints).
4. **System Architecture:** Flowcharts showing the Data Flow (DFD Level 0 and Level 1).
5. **Individual Contributions:** A table detailing exactly which module/form was implemented by which member.